

# Tan Dai Ngo

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## EDUCATION

**University of Victoria** | MEng, Applied Data Science (Co-op) *Victoria, BC | Sep 2025 - Dec 2027*

- **Coursework:** Optimization for Machine Learning, Systems for Massive Datasets. **GPA:** 8.78 / 9.00.

**University of Chicago** | MS, Computer Science *Chicago, IL, USA | Sep 2020 - Mar 2022*

- **Coursework:** Distributed Systems, Databases, Discrete Mathematics, Applied Software Engineering.

**University of Washington** | BS, Economics *Seattle, WA, USA | Sep 2015 - Aug 2019*

- **Minors:** Applied Mathematics, Informatics. **GPA:** 3.57 / 4.00.

## EXPERIENCE

**Software Developer** | T-Mobile (via BeaconFire Inc.) *Bellevue, WA, USA | Jun 2022 - Apr 2024*

- Automated reprocessing of rejected roaming tests with weekly Kafka retries, cutting manual effort by 80%.
- Owned REST endpoints on 5+ Java and Spring Boot roaming services, holding Cassandra reads under 1 second.
- Traced 20+ cross service defects to their root cause and shipped fixes via Jenkins across 50+ releases.
- Debugged production issues in Splunk and documented systems and runbooks so the work survived handover.

**Math and Computer Science Teacher** | APU American International Schools *Da Nang, Vietnam | Aug - Dec 2024*

- Taught Python, algorithms, object oriented design and software engineering fundamentals to 50+ students.
- Led coding projects spanning debugging, algorithmic reasoning and software performance optimization.

## PROJECTS

**Multi-Stage IoT Intrusion Detection** | Python, XGBoost, PyTorch *UVic Capstone, team of 7 | May - Aug 2026*

- Compared six supervised classifiers on 206,000 CIC IoT-DIAD 2024 flows with preprocessing fit on training only.
- Selected binary XGBoost under a 1% false positive budget, reaching 99.5% recall at 0.94% FPR and 0.989 PR-AUC.
- Designed a four condition holdout that exposed capture session leakage, dropping PR-AUC from 0.981 to 0.125.

**Ocean Data ML Platform with RAG** | Python, Spark, XGBoost, Qdrant, FastAPI *Personal Project | Feb - Mar 2026*

- Harmonized 10.8 million NOAA and Ocean Networks Canada records into Bronze, Silver and Gold layers with lineage.
- Validated the pipeline on air and water temperatures, cutting air RMSE by 48% while persistence won on water.
- Served semantic search behind FastAPI, Qdrant and Ollama at hit@k 0.9 and term recall 0.85 on held out queries.

**Fuel Blending Prediction Service** | Python, CatBoost, FastAPI, AWS *Shell.ai Hackathon 2025 | Jul 2025*

- Built and deployed multi output regression across 10 blend properties through FastAPI on Dockerized AWS EC2.
- Benchmarked CatBoost against XGBoost under 5 fold cross validation and shipped it at 0.64 MAPE against 1.29.
- Engineered entropy based mixture metrics and serialized preprocessing so training and inference stay consistent.

**Automated Crypto Trading Bot** | Python, Coinbase API, AWS, pandas *Personal Project | Feb - Jul 2025*

- Implemented a mean reversion engine that buys 1% dips and sells on 2% rises inside rolling 60 minute windows.
- Added a trend strategy that buys only when the 8 period EMA is above the 20 and price is above the 200 EMA.
- Backtested both on four months of polled data, then deployed to AWS EC2 with per market caps from a CSV config.

## TECHNICAL SKILLS

**Languages & Programming:** Python, Java, SQL, Go, JavaScript, Cypher, MATLAB, object oriented programming

**Machine Learning & Evaluation:** PyTorch, XGBoost, CatBoost, scikit-learn, pandas, NumPy, Optuna, MLflow

**Data, Retrieval & Messaging:** Spark, PySpark, ETL, Qdrant, RAG, embeddings, vector search, Cassandra, Kafka

**Backend, Cloud, CI/CD & Testing:** Spring Boot, FastAPI, REST, Docker, AWS (EC2, S3, Lambda), Jenkins, pytest

**Credentials:** Google Advanced Data Analytics (2025); Shell.ai Hackathon 2025; IMC 64 Bids, top 250 of 1,500